

| Call for Expressions of Interest                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <i>Date of Issue:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | September 2026                                                    |
| <i>Expected Completion Date:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | May 2027                                                          |
| <i>Project Title:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ARM Teaching Emulator                                             |
| <i>Company:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Please provide company name                                       |
| Contact Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                   |
| <i>Lead Contact:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Ian Knight                                                        |
| <i>Lead E-mail:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ian.knight2@nottingham.ac.uk                                      |
| <i>Additional Contact:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Additional contact name                                           |
| <i>E-mail:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Additional contact e-mail                                         |
| <i>Additional Contact:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Additional contact name                                           |
| <i>E-mail:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Additional contact e-mail                                         |
| <i>Additional Contact:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Additional contact name                                           |
| <i>E-mail:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Additional contact e-mail                                         |
| Additional Info                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                   |
| <i>Students must sign NDA:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <input type="checkbox"/>                                          |
| <i>Project uses personal data:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <input type="checkbox"/>                                          |
| <i>Custom hardware required:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <input type="checkbox"/><br>If Applicable: Please provide details |
| <i>Relevant research areas (if applicable)<sup>1</sup>:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Click or tap here to enter text.                                  |
| Project Background Information                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                   |
| <p><b>Assembly programming is taught in the first semester of our course, and we currently rely on an emulator program called KoMoDo which was last updated in 2015. An emulator is useful because it allows us to experiment with assembly programs without needing to run them on an ARM platform and work with an actual assembler and linker. This project will involve developing an up-to-date ARM emulator capable of supporting all the same functionality as KoMoDo, with improvements based on your own experiences.</b></p> <p><b>This project could help improve teaching of Assembly Programming, as well as my own Compilers module which also uses ARM.</b></p> |                                                                   |

---

<sup>1</sup> An overview of different research areas in the school of computer science can be found [here](#).

| Requirements                                                                                                                                                                                                                                                                                                                            |                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <p><i>The group must provide:</i></p> <p>A GUI application implementing an ARM emulator. At minimum the emulator should support the range of instructions covered in first year Assembly Programming, but a wider range of instructions would be ideal. The emulator should be able to read and execute a single ARM assembly file.</p> |                          |
| Skills                                                                                                                                                                                                                                                                                                                                  |                          |
| <p><i>Highly desirable:</i><br/>Understanding of ARM assembly<br/>GUI programming</p>                                                                                                                                                                                                                                                   | <p><i>Desirable:</i></p> |
| Submission / Award information                                                                                                                                                                                                                                                                                                          |                          |
| <i>Date of Submission:</i>                                                                                                                                                                                                                                                                                                              | September 2026           |
| <i>Date of Pitch:</i>                                                                                                                                                                                                                                                                                                                   | October 2026             |
| <i>Notification of award:</i>                                                                                                                                                                                                                                                                                                           | October 2026             |

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.